

How to Read Papers

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I. Where can I
find newly
published papers?



arxiv.org

The one stop shop for pre-prints,
post-prints, and everything!

- **Papers in 8 different categories:**
 - Physics, Mathematics, Computer Science, Quantitative Biology, Quantitative Finance, Statistics, Electrical Engineering & Systems Science, Economics
- **ADS (Astrophysics Data System)**
 - complements arXiv by also indexing published journal literature, not just preprints



ui.adsabs.harvard.edu

Where every astronomy paper
lives!

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II. How to start reading a paper?

Where to start depends on why you're there

- **Confirming it's the right paper?** Start with the Abstract
- **New to the field?** Start with the Introduction — it builds up the background you need
- **Chasing a specific result?** Start with the Figures — a fast, visual guide to what the paper found
- **Replicating their work?** Start with the Methods — your how-to guide
- **Pressed for time?** Start with the Conclusion/Summary, then jump to whatever stands out

III. How to
choose what
papers to read?

Three entry points

- **Motivated by an object?** Search it on [SIMBAD](#) — it lists every publication about or mentioning it
- **Motivated by a topic, idea, or claim?** AI tools can surface literature context — always ask them for arXiv or ADS links so you can verify
- **Want to vibe-read a subfield?** Search ADS/arXiv for "Review" papers by keyword — ARAA is a great source

IV. AI tools that help in reading papers

What AI tools are available?

- [Scholarcy](#)
- [SciSummary](#)
- [tldrthis.com](#)
- **Traditional AI tools**
 - I personally prefer Claude
 - ChatGPT works for things that rely solely on text and what's given, not necessarily for additional data driven analysis

V. Are there less
technical versions
of papers?



astrobites.org

Daily paper summary web-based
journal

- Daily paper summaries
- Reviews on field
- Interviews of people in field
- Conference coverages

VI. Book club for papers

Make it a group activity

- **Journal clubs** — most research institutions run one: think of it like a news forum, but discussing the latest papers instead of headlines
 - Tools like Benty Fields can help track and surface new papers for the group
 - No club at your institution? Start one — especially as an undergrad-oriented club under a senior researcher's guidance. Grad students and postdocs are often glad to help
- **Online clubs** — platforms like alphaXiv let you annotate papers and discuss them directly, Reddit-style
- **Microblogging** — Bluesky has become the go-to in the astro community (Twitter/X still exists, but most have migrated)

V. Assignment

Reading a paper at different levels

1. Go to astrobites.org and look for an article tagged “[daily paper summary](#)” for a topic you’re interested in, and read that for a beginner level explanation
2. Then read the open access version, discuss and ask each other questions– like an actual forum!
3. Make a summary slide that includes:
 - a. one plot
 - b. some info: the goal of the paper and what result it arrived at
 - c. The main new thing you learned